

Trainings ▾

Preparatory Steps

- Disconnect the air piping from the intake manifold.
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html)
- Remove the engine brakes, if applicable, from the engine. Refer to Procedure 020-024 in Section 20.
(/qs3/pubsys2/xml/en/procedures/35/35-020-024-tr.html)
- Remove the valve push tubes and injector push rods from the engine. Refer to Procedure 004-014 in Section 4.
(/qs3/pubsys2/xml/en/procedures/35/35-004-014-tr.html)

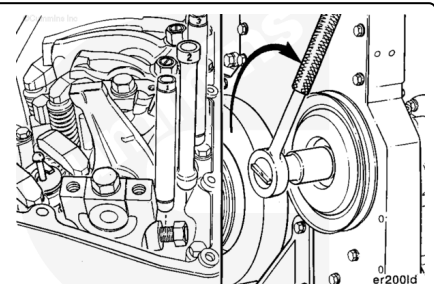


Remove

Rotate the rocker levers up.

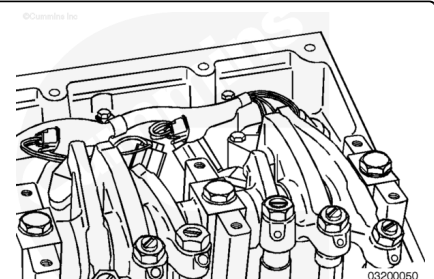
Mark the position of the crossheads as they are removed. They **must** be installed in the same position from which they were removed.

Remove the crossheads to prevent them from falling into the lower area of the engine.



The internal engine harness is located inside the rocker housing.

A pass through connector joins the internal and external engine harness. This connector has two clips that can manually be opened to allow separation of the two harnesses. Engines with pass through connectors in the front of the engine have clips on the outside, engines with pass through



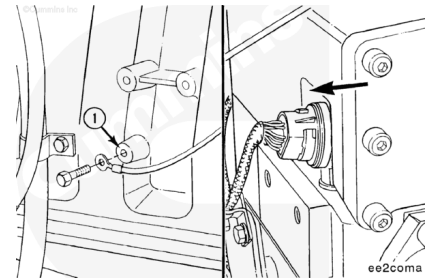
connections in the rear of the engine have clips on the inside of the rocker housing that require the valve cover to be removed to access.

Disconnect the ground wire connection of the actuator harness (1) from the engine.

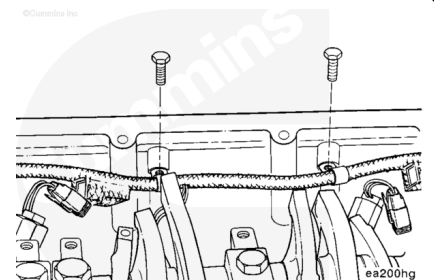
Tag the end for future identification.

Remove the external section of the actuator harness.

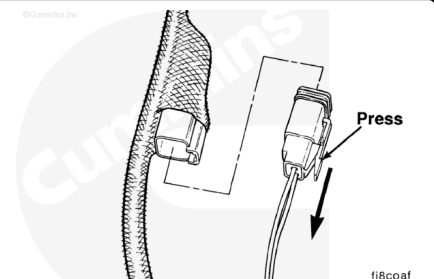
Use care **not** to damage the plug as it is being removed from the rocker housing.



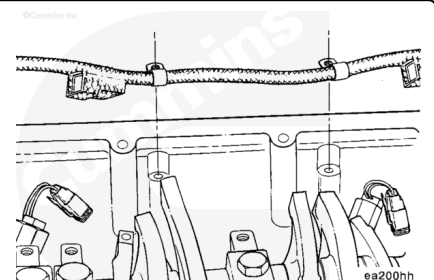
Remove the internal engine harness holding clamp capscrews.



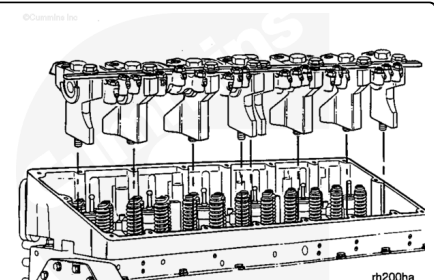
Separate the Deutsch™ 2-pin connectors from the internal engine harness to the fuel injectors.



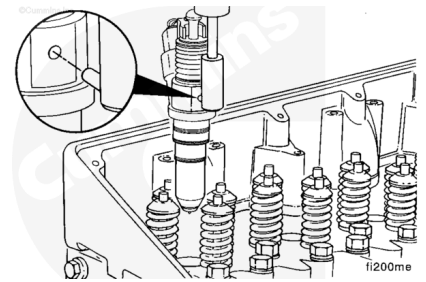
Remove the internal engine harness from the engine.



Remove the rocker lever assemblies. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/35/35-003-009-tr.html>)



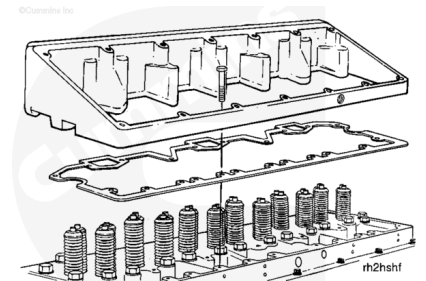
Remove the injectors. Refer to Procedure 006-026 in Section 6. ([/qs3/pubsys2/xml/en/procedures/35/35-006-026-tr.html](http://qs3/pubsys2/xml/en/procedures/35/35-006-026-tr.html))



Remove the rocker housing mounting capscrews.

Remove the rocker housing and the intake manifold as a complete unit.

Note : If high blowby is a symptom, the rocker housing gasket must be inspected before it is removed from the cylinder head.



Clean and Inspect for Reuse

⚠ WARNING ⚠

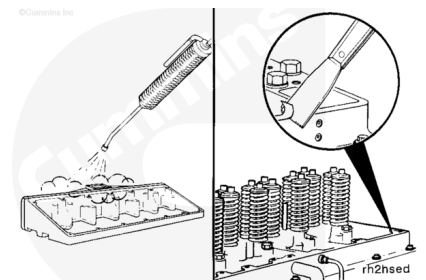
When using a steam cleaner, wear safety glasses or a face shield as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use steam to clean the rocker lever housing. Dry with compressed air.

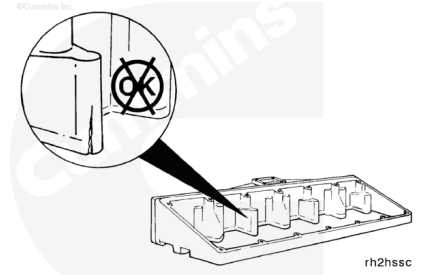
Clean the cylinder head gasket surface.



Inspect the rocker housing gasket for cuts or damage on the portion of the gasket that seals around the intake ports. Damage to the gasket can cause high blowby.

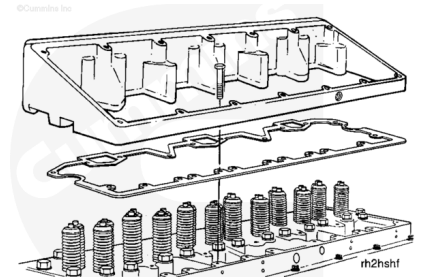
Inspect the housing for cracks or damage.

If the housing is cracked, it **must** be replaced.

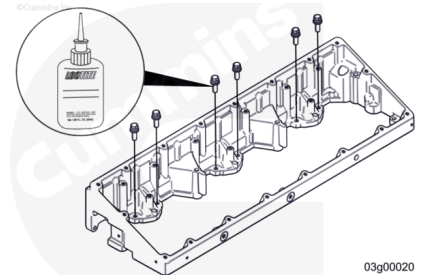


Install

Use a new gasket and install the rocker lever housing.



Note : Oil resistant, medium strength thread locker, such as Loctite™ 243, or equivalent, **must** be used when assembling the six short capscrews located on both sides of the three intake ports.



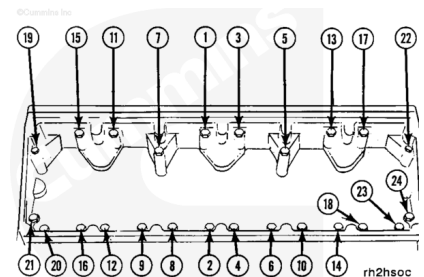
Step 1: Snug all bolts.

Step 2: Tighten the 24 capscrews in the sequence shown.

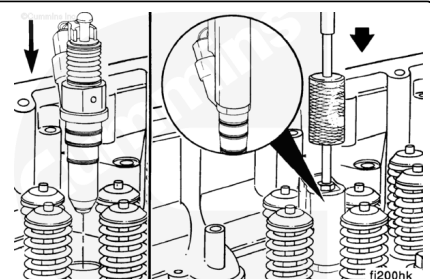
Torque Value: 34 n•m [25 ft-lb]

Step 3: Tighten the 24 capscrews in the sequence shown.

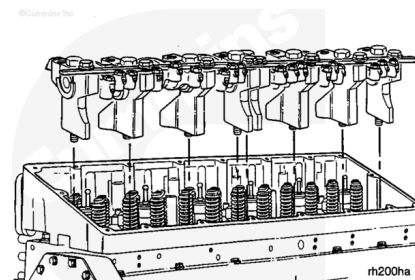
Torque Value: 61 n•m [45 ft-lb]



Install the injectors. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/35/35-006-026-tr.html>)

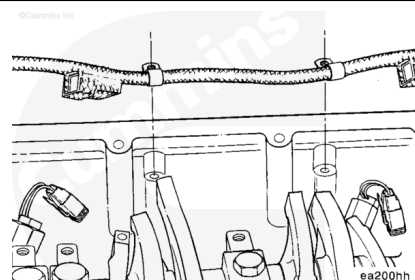


Install the rocker lever assemblies. Refer to Procedure 003-009 in Section 3. (/qs3/pubsys2/xml/en/procedures/35/35-003-009-tr.html)



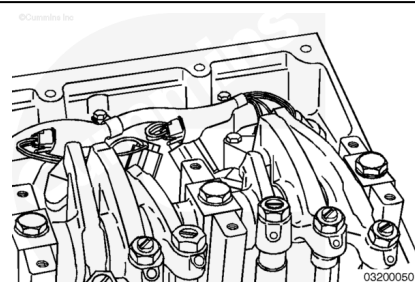
Install the internal engine wiring harness on the engine.

The harness and connectors **must** be clear of any moving parts, so the wires will **not** be damaged during engine operation.



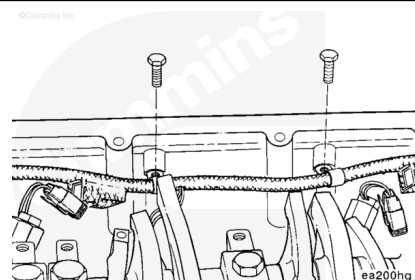
Connect the 15-pin connector of the internal engine harness to the 15-pin connector of the external engine harness.

Note : The internal and external Metri-Pack™ connectors are "keyed." Make sure both connectors are correctly aligned when they are joined together.

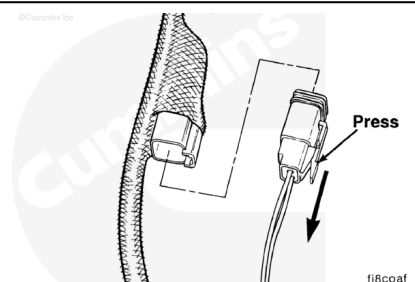


Install the holding clamps for the internal engine harness. Tighten the capscrews.

Torque Value: 20 n•m [15 ft-lb]



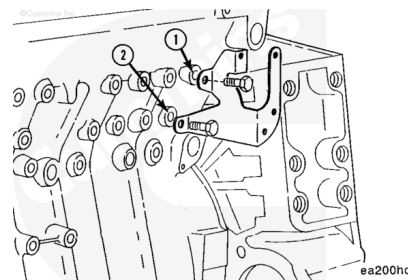
Connect the Deutsch™ 2-pin connectors from the internal engine harness to the fuel injectors.



Install the harness wire bracket to the cylinder block at points (1) and (2).

Install two mounting capscrews and tighten.

Torque Value: 41 n•m [30 ft-lb]

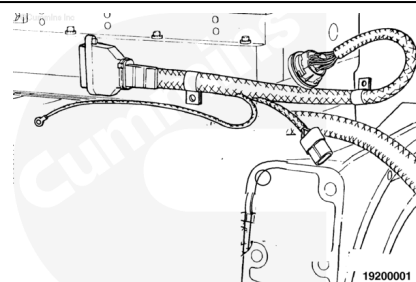


Install the external actuator harness on the engine.

Install the harness holding clamps to the support bracket and to the rear of the engine block.

Tighten the capscrew.

Torque Value: 20 n•m [15 ft-lb]



Install the oil temperature sensor (1) and the oil pressure sensor (2) to the cylinder block.

Torque Value:

Temperature Sensor

1. 34 n•m [25 ft-lb]

Torque Value:

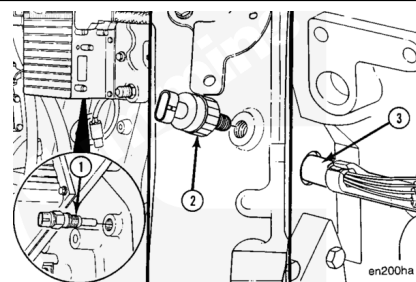
Pressure Sensor

1. 30 n•m [25 ft-lb]

Install the engine position sensor (3) to the rear of the gear housing.

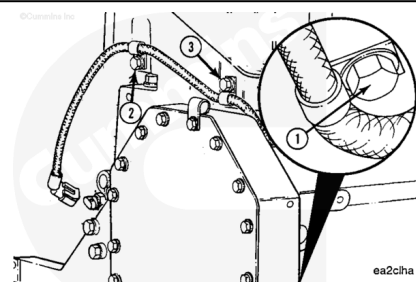
Torque Value:

1. 34 n•m [25 ft-lb]



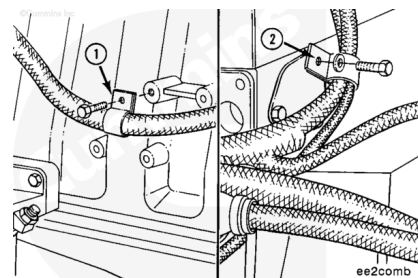
Install the sensor wire harness.

Use wire clamps to fasten the coolant temperature sensor lead to the gear housing (1) and to the front of the rocker lever housing (2) and (3).



Install a wire clamp around the sensor harness and actuator fuel shutoff solenoid wire, and fasten the clamp to the cylinder block at point (1).

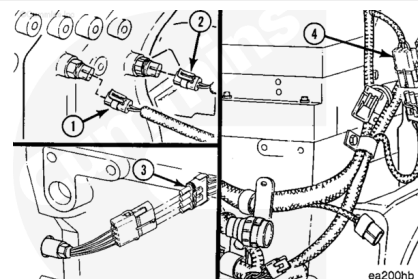
Install another wire clamp around the intake temperature and boost pressure sensor wires and actuator harness, and fasten them to the camshaft rear cover plate (2).



Note : Care must be used when matching connectors with sensors. Each sensor has its own unique connection.

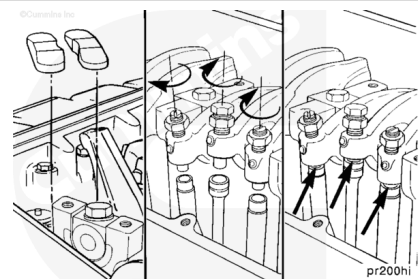
Note : Connect the correct sensor harness wire connectors to the following sensors:

- Oil temperature sensor (1)
- Oil pressure sensor (2)
- Engine position sensor (3).



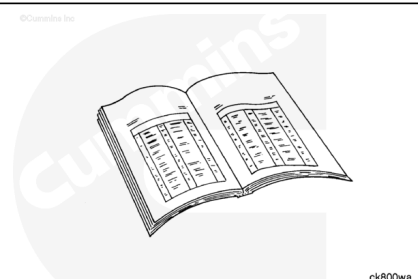
Connect the sensor and actuator harnesses together with Weather Pack™ connector (4).

Install the crossheads in the same position from which they were removed , and rotate the rocker levers down.



Finishing Steps

- Adjust the valves and injectors. Refer to Procedure 003-004 in Section 3.
(/qs3/pubsys2/xml/en/procedures/35/35-003-004-tr.html)
- Install the push rods and push tubes. Refer to Procedure 004-014 in Section 4.
(/qs3/pubsys2/xml/en/procedures/35/35-004-014-tr.html)
- Install the engine brakes, if applicable. Refer to Procedure 020-024 in Section 20.



(/qs3/pubsys2/xml/en/procedures/35/35-020-024-tr.html)

- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html)
- Connect the air piping to the intake manifold. Refer to the manufacturer's specifications for the correct torque value.
- Operate the engine and check for leaks.

Last Modified: 22-Sep-2013
